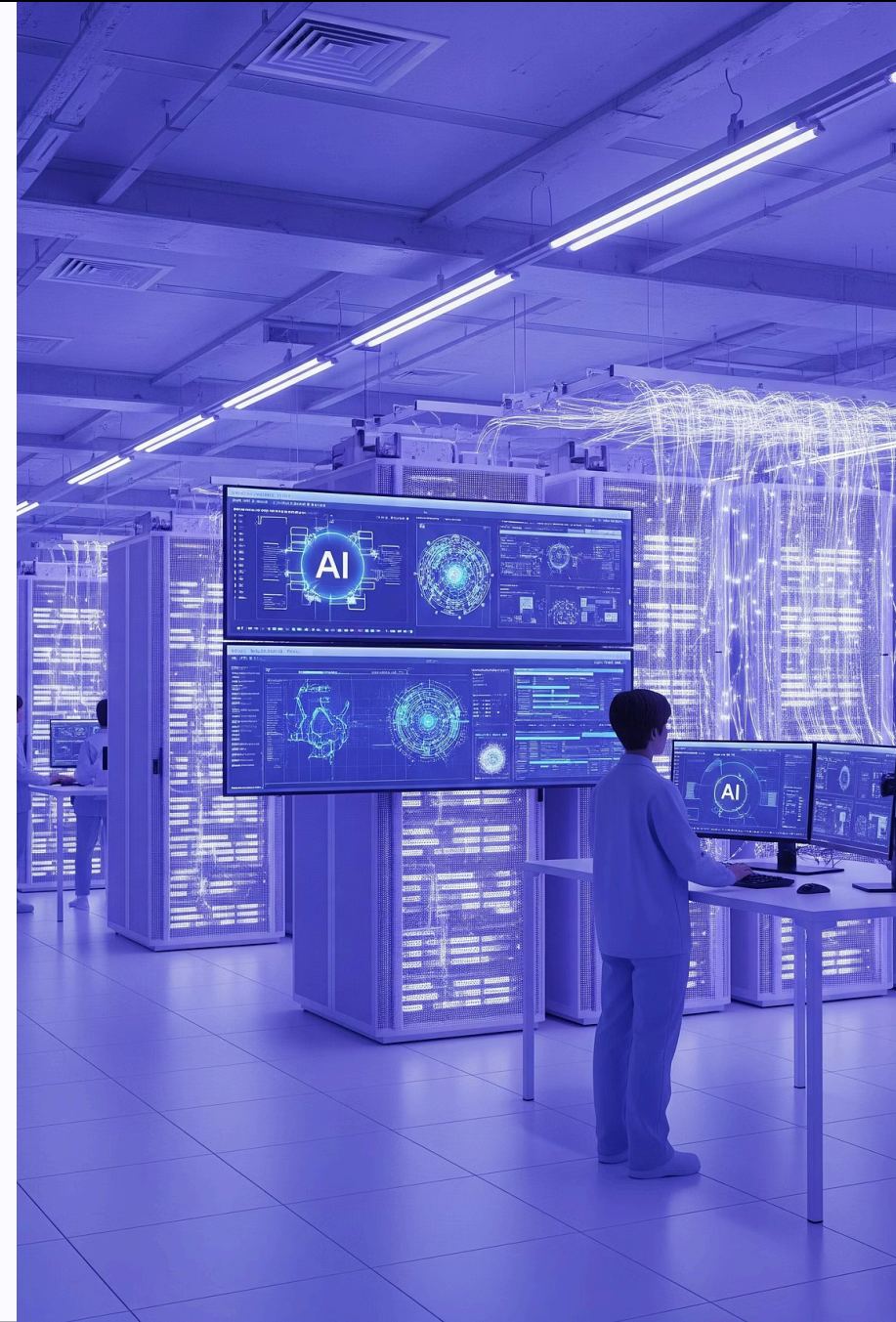


What Is MLOps and CI/CD for AI?

A practical introduction to the infrastructure and practices that power production AI systems at scale.



Why MLOps Matters

ML Models Are Dynamic

Unlike traditional software, AI models require continuous updates to maintain accuracy as data changes.

Traditional DevOps Falls Short

AI workflows introduce unique challenges around data, experimentation, and model behavior that standard DevOps can't address.

MLOps Bridges the Gap

Bringing DevOps principles into ML development creates a foundation for **reliable, reproducible, and scalable** AI systems.



MLOps is now **critical infrastructure** for organizations serious about delivering AI value in production.

What Is MLOps?

MLOps combines Machine Learning with DevOps practices to manage the complete ML lifecycle from data to deployment.

Data Engineering
Collecting, cleaning, and versioning data

Automation
Creating CI/CD pipelines for the entire process



Model Development

Training, experimenting, and evaluating models

Deployment

Packaging and serving models in production

Monitoring

Observing performance and detecting drift

The ultimate goal: Ensure AI models consistently deliver business value in production

Core Pillars of MLOps

Data Management

- Data versioning & lineage
- Feature stores
- Data quality validation
- ETL pipelines

Model Training & Tracking

- Experiment tracking
- Hyperparameter optimization
- Model versioning
- Reproducible environments

Deployment & Serving

- CI/CD automation
- Container orchestration
- Model registry
- Inference optimization

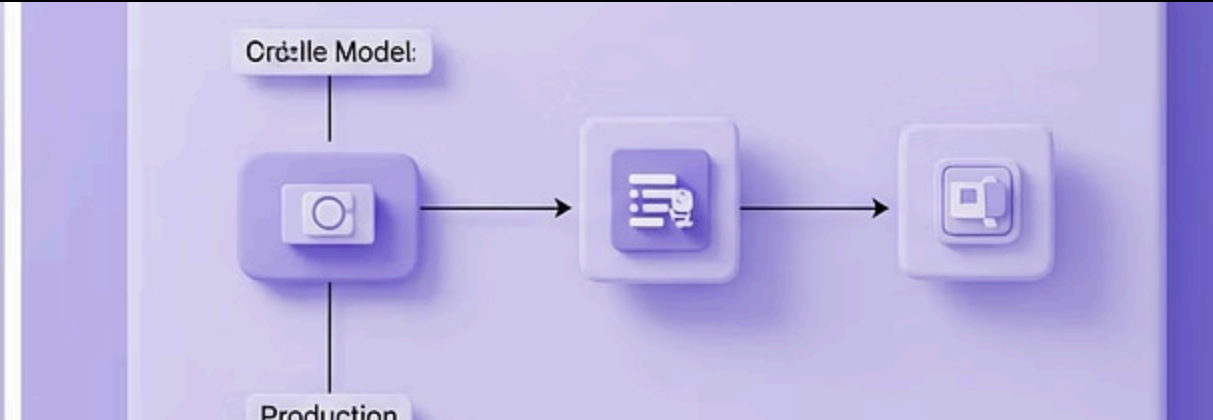
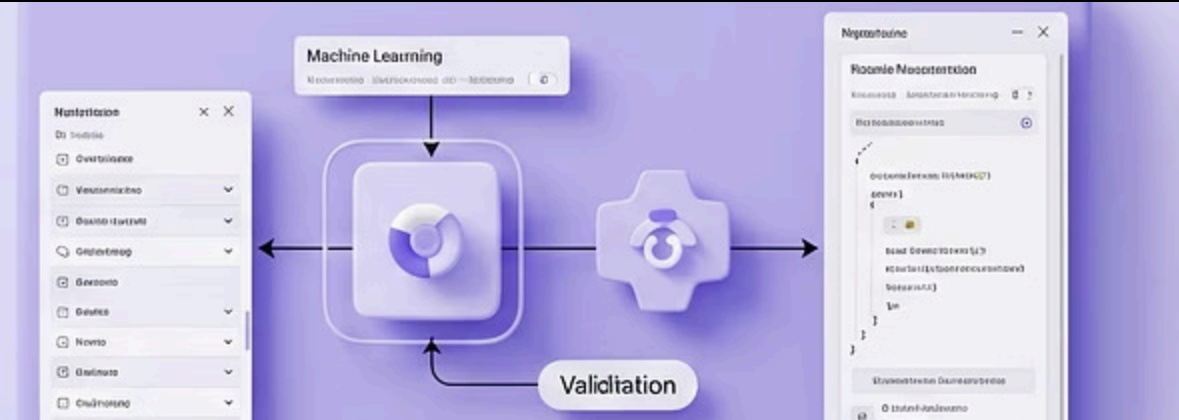
Monitoring & Governance

- Data & model drift detection
- Performance metrics
- Compliance & auditability
- Automated alerting

Collaboration

- Cross-functional workflows
- Knowledge sharing
- Role-based access
- Documentation

Successful MLOps requires tight integration between data scientists, ML engineers, and operations teams.



What Is CI/CD in AI?

Continuous Integration (CI)

Automatically test and validate changes to:

- ML code & algorithms
- Data preprocessing
- Model performance
- Integration with other systems

Continuous Delivery/Deployment (CD)

Automate the release process for:

- Model artifacts & weights
- Serving infrastructure
- Configuration updates
- A/B testing new models

CI/CD for AI ensures models stay **fresh, reliable, and valuable** through automated training and deployment cycles.

CI/CD Workflow for ML



Code Commit

Developer pushes changes to ML code, configs, or pipeline definitions



Data Pipeline

Validate data quality, preprocess features, generate training datasets



Model Training

Train model, track metrics, compare to baseline performance



Testing

Run unit, integration, and model-specific tests



Registry

Register approved model version with metadata



Deployment

Deploy to staging then production with monitoring

 Modern CI/CD pipelines for ML can conditionally retrain models only when significant data or code changes occur, saving compute resources.

Tools for CI/CD in AI

Pipeline Automation



- GitHub Actions
- GitLab CI
- Jenkins
- CircleCI

Experiment & Versioning



- MLflow
- DVC (Data Version Control)
- Weights & Biases
- Neptune.ai

Workflow Orchestration



- Kubeflow Pipelines
- Apache Airflow
- Prefect
- Metaflow

Deployment Infrastructure



- Docker
- Kubernetes
- Seldon Core
- KServe

Cloud-Native Options



- AWS SageMaker Pipelines
- Google Vertex AI
- Azure ML
- Databricks MLflow

Benefits of MLOps + CI/CD



Accelerated Innovation

Reduce time from idea to production from months to days with automated pipelines

Reliability at Scale

Consistent, repeatable deployments with built-in safeguards against performance degradation

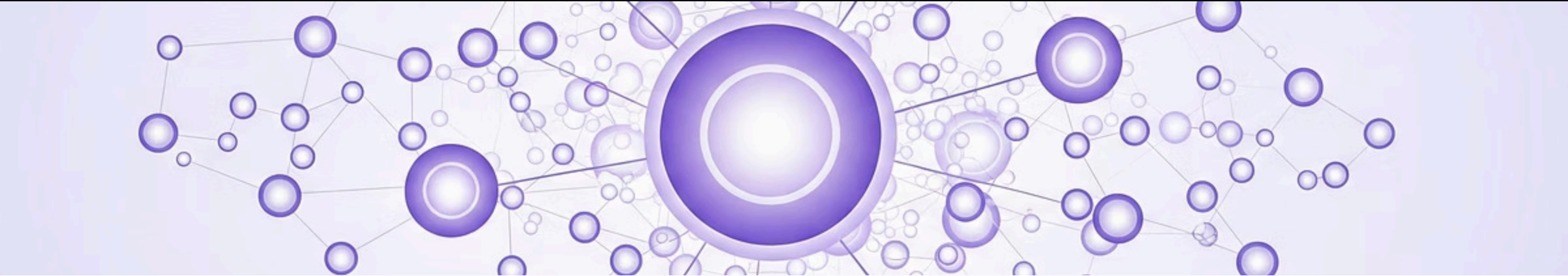
Cross-Functional Collaboration

Bridge the gap between data science and engineering with shared tooling and workflows

Continuous Value Delivery

Keep models fresh and responsive to changing data patterns and business needs

Organizations with mature MLOps practices deploy models 5-10x more frequently with significantly fewer production issues.



Challenges in AI CI/CD

1

Data Dependency Complexity

Models depend on ever-changing data, not just code. Pipeline triggers must account for both code and data changes.

2

Resource-Intensive Training

Large models and datasets mean long training times. CI/CD must balance frequency, cost, and compute resources.

3

Testing Complexity

ML systems require statistical evaluation beyond traditional unit tests. Need to validate both code *and* model behavior.

4

Organizational Alignment

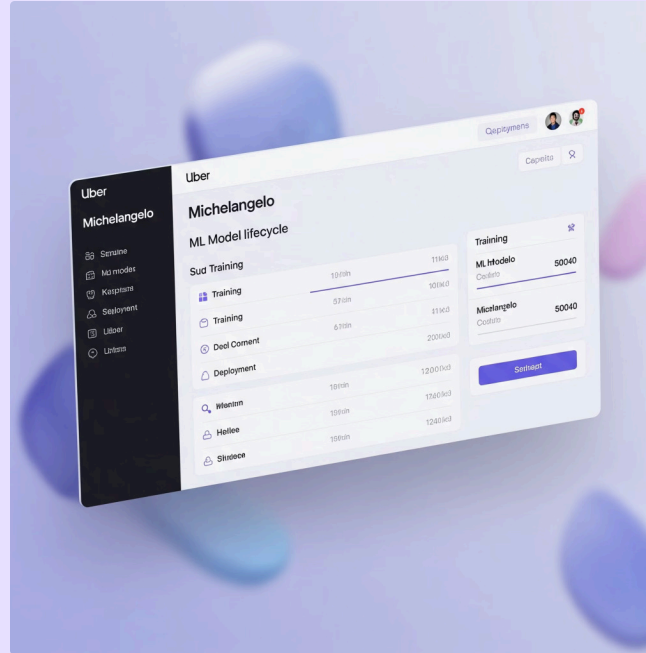
Requires cultural shift from research-oriented data science to engineering-driven practices with shared responsibility.

Real-World Examples



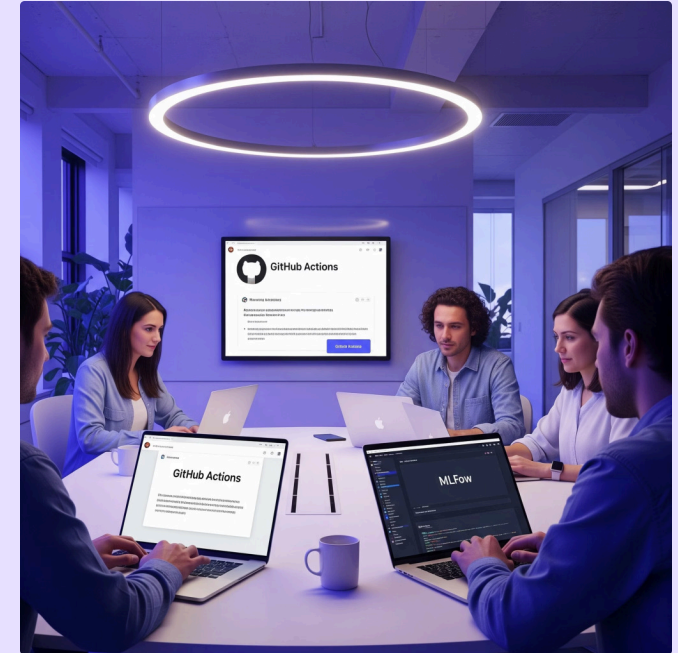
Netflix

Automated CI/CD pipelines for recommendation models, updating multiple times daily while maintaining quality



Uber (Michelangelo)

End-to-end ML platform managing thousands of models across forecasting, pricing, and matching



Startups

Leveraging GitHub Actions + MLflow + Docker for lightweight but effective MLOps without massive infrastructure

CI/CD has become the backbone of production AI at all scales, from startups to tech giants

Key Takeaways

MLOps Is Essential

Applying DevOps principles to the ML lifecycle is now critical for any organization deploying AI at scale

Tools Are Maturing

MLflow, Kubeflow, GitHub Actions, Docker, and Kubernetes form the foundation of modern ML infrastructure

CI/CD Drives Success

Automation for integration, testing, and deployment enables reliable, consistent model updates



MLOps and CI/CD for AI are now must-have skills for engineers working on production machine learning systems